

Research Essentials

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Sydney Informatics Hub
Core Research Facilities

sydney.edu.au/sydney-informatics-hub



THE UNIVERSITY OF
SYDNEY

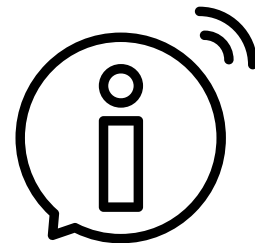


Slides available here

CRICOS 00026A TEQSA PRV12057



Acknowledging SIH



- All University of Sydney resources are available to researchers free of charge. The use of the SIH services including the Artemis HPC and associated support and training warrants acknowledgement in any publications, conference proceedings or posters describing work facilitated by these services.
- The continued acknowledgment of the use of SIH facilities ensures the sustainability of our services.

Suggested wording for use of workshops and workflows:

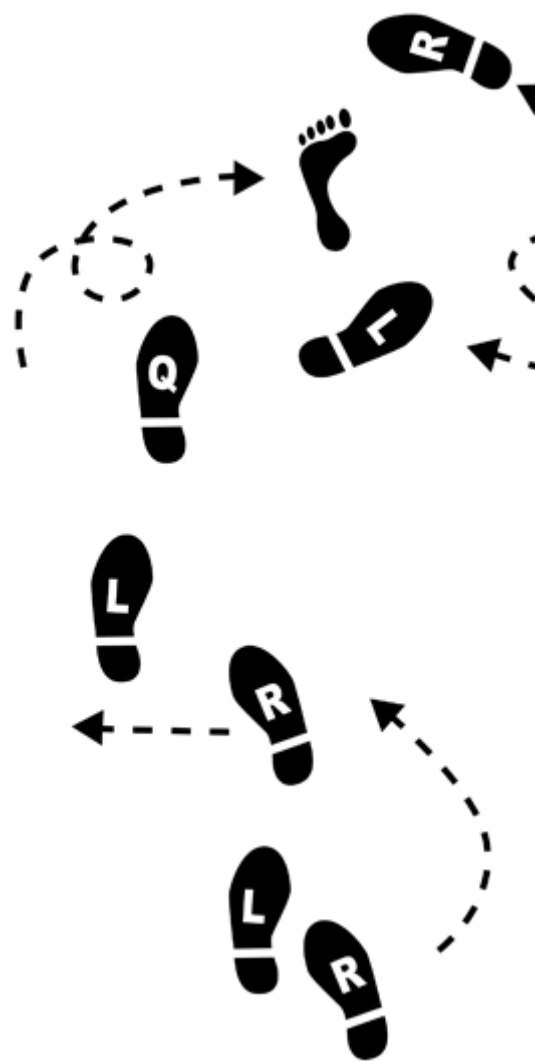
- *“The authors acknowledge the Statistical workshops and workflows provided by the Sydney Informatics Hub, a Core Research Facility of the University of Sydney.”*

What is a workflow?

- Every statistical analysis is different, but all follow similar paths. It can be useful to know what these paths are.
- We have developed practical, step-by-step instructions that we call ‘workflows’, that you can follow and apply to your research.
- We have a general research workflow that you can follow from hypothesis generation to publication.
- And statistical workflows that focus on each major step along the way (e.g. experimental design, power calculation, model building, analysis using linear models/survival/multivariate/survey methods).

Statistical workflows

- Our **statistical workflows** can be found within our workshop slides.
- **Statistical workflows** are software agnostic, in that they can be applied using any statistical software.
- To access these statistical workflows and more, visit our [Workshops and Workflows](#) page.



Software workflows



- There may also be accompanying **software workflows** that show you how to perform the statistical workflow using particular software packages (e.g. R or SPSS). We won't be going through these in detail during the workshop. If you are having trouble using them, we suggest you attend our monthly [Hacky Hour](#) where SIH staff can help you.
- Our software workflows contain:
 - o R code and comments.
 - o SPSS syntax as well as screenshots of the point and click procedures and written methods.
 - o Screenshots of the point and click procedures and written methods for other bespoke software.

During the workshop



- Ask short questions or clarifications during the workshop (either by Zoom chat or verbally). There will be breaks during the workshop for longer questions.



- Slides with this blackboard icon are mainly for your reference, and the material will not be discussed during the workshop.



- Challenge questions will be encountered throughout the workshop.

Research Essentials workshop overview

8-step general research workflow and other resources

- Where does this Workshop fit into the research process?
- Where does it fit in with other SIH training and support on offer?

Setting up your data for most analyses:

- Step 3: Collect and store data
- Step 4: Cleaning data

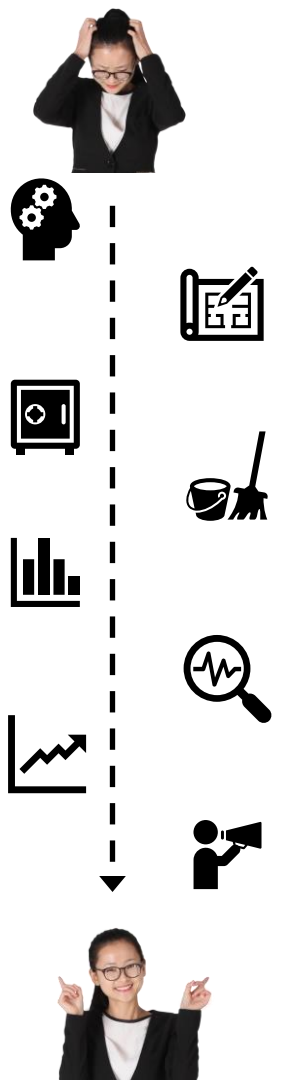
Workflow examples for common analyses – brief introduction to:

- Step 5: Exploratory data analysis
- Step 6: Inferential analysis

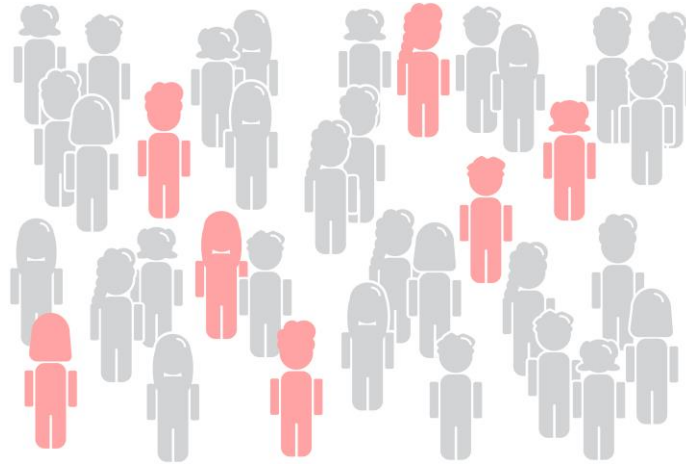
8-step general research workflow

General research workflow

1. **Hypothesis Generation** (Research/Desktop Review)
2. **Experimental and Analytical Design** (Sampling, power, ethics approval)
3. **Collect/Store Data**
4. **Data cleaning**
5. **Exploratory Data Analysis (EDA)**
6. **Data Analysis aka inferential analysis**
7. **Predictive modelling**
8. **Publication**



6. Statistical Inferential analysis – from sample to population



Statistical inference:

“The theory, methods, and practice of forming judgments about the parameters of a population, usually on the basis of random sampling.”

Collins English Dictionary

7. Predictive modelling – Inferential predictive statistics versus machine learning predictive analytics

Inferential predictive statistics



Predicting temperature rise in climate change.

Machine learning/predictive analytics



Very accurately verify fingerprint to unlock a mobile phone.



Ecosystem of SIH statistical training

Workflow step	Other training
1. Hypothesis generation	<u>Research Essentials</u>
2. Study design	<u>Experimental Design</u> <u>Power and Sample Size</u> <u>Design and Analysis of Surveys 1</u> + <u>Design and Analysis of Surveys 2: Advanced Topics</u> <u>Statistical Model Building</u>
3. Collect/store data	<u>Research Essentials</u>
4. Data cleaning	<u>Research Essentials</u>
5. Exploratory data analysis	<u>Research Essentials</u>
6. Inferential analysis	<u>Linear Models 1 to 3</u> + <u>Statistical Model Building</u> <u>Introduction to Survival Analysis</u> <u>Meta-Analysis: An Introduction</u> <u>Design and Analysis of Surveys 1</u> + <u>Design and Analysis of Surveys 2: Advanced Topics</u> <u>Multivariate Statistical Analysis 1: Dimension Reduction</u>
7. Predictive modelling	[Predictive analytics: <i>Introduction to machine learning in R/Python (SIH Data Science)</i>]: <u>From 26th August</u>
8. Publication	



SIH training

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Different offerings, in person, online, and hybrid content in a variety of formats from webinars to interactive workshops.



Attendees at all career levels, from undergraduate students to senior professors, and representation from every Faculty and School.



Partnerships with national organisations like Australian BioCommons:
biocommons.org.au/training-cooperative.



Find out more on our training calendar:
sydney.edu.au/informatics-hub/training. Or stay up to date with our newsletter.



Statistics	Data Science	Research Computing	Bioinformatics	Events
Fundamentals	Machine Learning	High Performance Computing	'omics Techniques	Hacky Hour
Modelling	Visualisation	Cloud Computing	Reproducible Pipelines	Summer Schools
Specialist	Natural Language Processing	Containers	Data Analytics	Coding Challenges
	Geospatial Analysis	Workflows	National Compute Infrastructure	



Ecosystem of other USyd training

Workflow step	Other training
1. Hypothesis generation	Library research support: Literature and systematic review
2. Study design	
3. Collect/store data	RedCap- Various trainings for survey data from introduction to advanced Research Data Management modules and techniques
4. Data cleaning	
5. Exploratory data analysis	
6. Inferential analysis	
7. Predictive modelling	
8. Publication	Library research support: Publishing



Research data management

- eNotebook
- RedCap
- Research data store
- OneDrive (Office 365)

Supported
platforms



- Google Drive
- Survey Monkey
- Portable Drives e.g. USB

Do not
use



[University-supported research data platforms: Features and comparisons](#)

Research data that is managed optimally improves research efficiency and reach, as well as ensuring its integrity and security, and meeting legislative/policy/funding/publishing requirements.

The Research Data Consulting team assists researchers to enhance their research productivity and improve data management practices. They provide:

- Short consultations to integrate digital tools and data management into your research.
- Training and functional support for university supported tools/platforms.

Research computing



Institutional



National



Commercial



More info

The University of Sydney

Research Problems

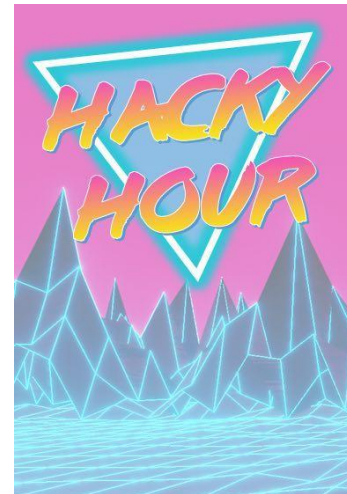
Research Insights

Hacky Hour

For researchers who code or analyse data



- A monthly meetup where anyone from the University – students, staff and university affiliates – can collaborate and get support e.g., swap notes, get help, or learn new techniques in programming and data science.
- Experts & mentors from SIH and across the University will be available to advise and answer questions on coding, data analytics or digital tools.
- Come join us on zoom the **3rd Wednesday of every month, 2 - 3pm!**
- sydney.edu.au/informatics-hub/hacky-hour



How to engage with us



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Research and prototype foundry

Sydney Analytical

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Workshops and training

Research project support

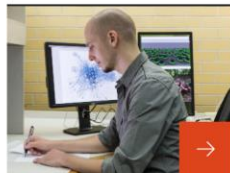
Sydney Informatics Hub

Enabling excellence in data and compute intensive research

We provide support, training, and expertise in research data management, statistics, data science, software engineering, simulation, visualisation, bioinformatics, and research computing.



Workshops and training



Research project support

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Request a quote or estimate for your research project or grant



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Contact

And fill out our request form

Or send us an email:

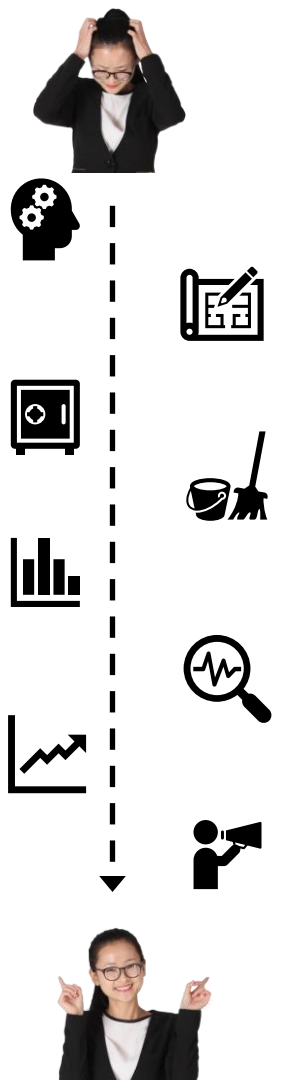


sih.info@sydney.edu.au



General research workflow

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The first question: Which car will you take?

Getting from step 3 to step 8 will involve using software. Will it be:

Graphical User Interface (GUI) – like an automatic car
Interactive, point and click
Easier to get started



Command line interface (CLI) – like a manual car
Writing code
Easier to handle complex and/or large data sets

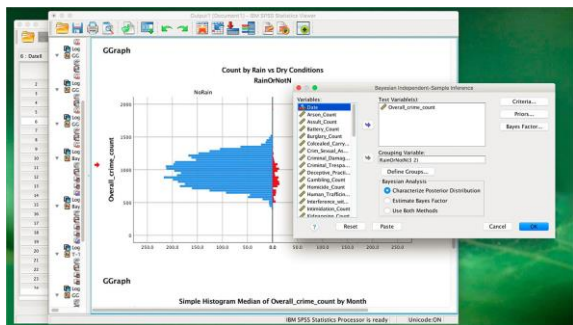
CLI



Software choice as the first question: GUI versus CLI

- Which software are you more familiar or comfortable with?
- How do you record your analysis for reproducible research?
- By documenting, you should always be able to re-run your analysis from start to finish (and get the same result)!
- If using interactive processing, you should keep track of the commands you run.

Graphical User Interface
(GUI) **SPSS**



Command line interface
(CLI) **R code**

```
R
R> lm(Sepal.Length ~ Species, data = iris)

Call:
lm(formula = Sepal.Length ~ Species, data = iris)

Coefficients:
(Intercept)  Speciesversicolor  Speciesvirginica
      5.006             0.930             1.582

R> |
```

3. Collect/store your data

A. Research data management

B. Organise your data for input into statistical software

A. Research data management

- **Data storage**
- Back up EVERYTHING including original data collection forms or raw data (images, electrical signals, DNA sequences, whatever).
- **Data entry - will you be using manual data entry?**
- Ideally double-data entry followed by comparison.
- Be wary of spreadsheets – especially entering, editing analysing in the same sheet.
- Statistical software generally doesn't allow easy editing once you have entered your data.



A. Research data management

- **Have you got a Research data management plan according to University policy?**
 - What is a Research Data Management Plan (RDMP)?
 - What are the university supported tools for data collection and storage?
 - What is an eNotebook?
 - Where can I store my data?
- **Consider appropriate folder/directory structure, file naming and version control for your project, *or at least your part of it.***
 - “Good enough practices for scientific computing”



A. Research data management

Guide to storing and managing your projects research data

University supported and licenced platforms

Platform/Tool	eNotebook	REDCap	Research Data Store (RDS)	OneDrive (Enterprise)	Teams (Enterprise)	Highly Protected SharePoint (Enterprise)	Australian Imaging Service (AIS)	Local storage, USB Drive	Other cloud tools (e.g. Google Drive, Dropbox)
function	electronic notebook	survey and data capture, including Clinical trials	networked data storage, large files, HPC access	cloud storage	chat, collaboration, cloud storage	collaboration, cloud storage	imaging repository and analytics	removable media, local storage	cloud storage
suitable for data classification	●●●	●●●	●●●	○●●	○●●	●●●	●●●	●	●
stored in Australia	✓	✓	✓	✓	✓	✓	✓	various	✗
external collaborator access	✓	✓	✓	✓	✓	✓	✓	✗	✗
context and commentary supported	✓	✗	✗	✓	✗	✗	n/a	✗	✗
syncing with local copy	n/a	n/a	n/a	✓	✓	✓	n/a	✗	✗
available storage	unlimited	unlimited	unlimited (default 2TB)	5TB	2TB+	25TB max (default 2TB)	unlimited	✗	✗
backup and disaster recovery	✓	✓	✓	✓	✓	✓	✓	✗	✗
audit trail/version control	✓	✓	✓	✓	✓	✓	✓	✗	✗
versioning retained	✓	manual	up to 60 days	7 years	7 years	7 years	✗	✗	✗

Version 6.0
December 2022
Endorsed by CIO



●	highly protected
○	highly protected data needs additional file encryption
●	protected
●	public

Highly Protected data may require additional encryption depending on some platforms. Protected data may benefit from encryption.

For more information about research data classifications, go to <https://sydney.edu.au/research-data-classifications>

For research data management enquiries, please contact digital.research@sydney.edu.au

Version control – keeping track of your files

- Use a separate directory for each discrete analysis.
- When processing data and intermediate files save with a new name.
 - Use a good file naming convention to keep track of files.
- Save frequently so if you lose a version, you do not have to redo too much work.
- For collaborative research consider using eNotebook or other version control systems, e.g. Git (free) or GitHub.
- Create a log file in the same directory and use version control (e.g. name sequentially, date/time stamp, for example:
 - “20230208 stats101 workshop v2.0.xlsx” (orders files chronologically)

Version control – keeping track of your files

Example of a version log file in Excel:

File name	Description	# Obs	# Vars
Mydata_v1_30102022.csv	Original data entry by KS, 1 record per person	250	34
Mydata_v2_01112022.csv	Eligible records only based on study inclusion criteria with new variables created for analysis: BMI calculated from recorded height and weight; babies age processed to be consistently in months instead of days and weeks as well; number of pets categorised (none, 1-2, 3+)	204	37

Data formats – tidy data

- Depending on the design of your experiment/survey you may have a mix of demographic data on each individual, and measurements
 - You may need multiple tables and a unique ID for each individual to link them, or just have the demographic data repeated when transforming to long format
- Wide and long can become relative terms especially if you have clusters of subjects
- Tidy data is an absolute term, which describes data transformed to:
 - One variable in each column
 - One observation per row
 - One value per cell

country	year	cases	population
Afghanistan	1999	1845	15467071
Afghanistan	2000	2666	20335360
Brazil	1999	3737	17206362
Brazil	2000	8488	17404898
China	1999	21258	1272015272
China	2000	21766	128053583

variables

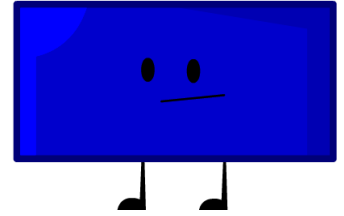
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observations

country	year	cases	population
Afghanistan	1999	1845	15467071
Afghanistan	2000	2666	20335360
Brazil	1999	3737	17206362
Brazil	2000	8488	17404898
China	1999	21258	1272015272
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values

B. Organising a dataset for analysis



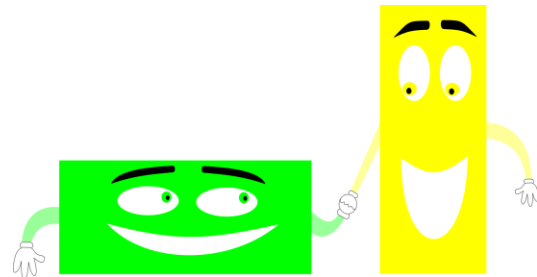
- **Most programs read in data in a rectangular format:**
 - E.g. A text file – you can read it in Notepad or any text editor or Excel, csv etc.
 - A header including column names in the first row.
 - Each row thereafter being the data itself (often corresponding to a single unit of interest – e.g. person, animal, plant, plot, farm, machine, business, school, hospital etc).
 - Each column represents one variable.
 - ID variable – identifies the subject.
 - Demographic variable – characteristics of the subject including their treatment.
 - Measurement variable – some observation on the subject.
 - A delimiter between each column (comma .csv and tab .tsv/.tab/.txt).

Pitfalls when coming from Excel

- Watch out for:
 - Merged cells
 - Cell comments
 - Colour coding
 - Blank rows
 - Data in multiple sheets
 - Particular coding of missing data/blanks/non-applicable
- Deal with the above in Excel before exporting to text. Sometimes these have been added to annotate the data, or make it easier to read. Other times, they are *part* of the data and must be represented some way in a text file.
- A good summary of these pitfalls is provided in [this paper](#).
- Check your data once it is imported into the statistical software.



B. Data formats – transformations



Patient ID	Time 1	Time 2	Time 3
1	50	55	60
2	47	49	50
...	...	...	...

**Wide/unstacked
format**



Patient ID	Time	Body weight
1	1	50
1	2	55
1	3	60
2	1	47
2	2	49
2	3	50
...	...	...

**Long/stacked
format**

B. Organising a dataset for statistical analysis – data coding

- Specify type of variable: ensure your analysis software knows whether a variable is continuous (numeric), categorical/factor/string/character (text).
- Label variables, either within the software or by keeping your own record (e.g. Age = Age at interview in years).
- Label variable values/'levels' within categorical variables, e.g. 1 = "Male", 2="Female", 3="Non-binary".
- Correctly code missing values according to software program: ensure your analysis software knows that the data is missing and not '0' or some other value (e.g. 999, 777, 99, 77).

4. Data cleaning



Data cleaning: data wrangling and data dictionary

A simple example of a data dictionary:

	A	C	D
1			
2	Questions	Categories	Code used
3	Q1_Age(years)	20-30	1
4		31-40	2
5		41-50	3
6		51-60	4
7		>60	5
8	Q2_Gender	male	1
9		female	2

- Data cleaning involves examining* the variables in the dataset and creating new variables for analysis by recoding/processing variables as required.
- Use short but informative variable names; it's a good idea to have a data dictionary.
- Names should keep track of transformations/recoding, e.g.
 - age = original data in years
 - age_c2 = age categorised into two categories (young vs old)

* More on how to examine, i.e. describe the distribution of variables later in this workshop.

Keep track of analyses

- **Remember you should be able to repeat analysis from the start, to demonstrate/enable reproducibility.**
- **For statistical programming languages:**
 - Name the program file logically.
 - Use structure, work in blocks or ‘chunks’ of code for different sections, e.g. ‘descriptive analyses’ – do it for all predictors in one go.
 - Log file – same name as program file, different extension – VERY important as record for interactive mode!
 - Use functions to avoid repetition.
 - Use appropriate level of comments, e.g. key steps and results.
 - Consider using an Rmarkdown/Quarto notebook if using R.
- Also covered in [Good enough practices in scientific computing](#).

Example R Markdown and Quarto files

Files Plots Packages Help Viewer Jobs

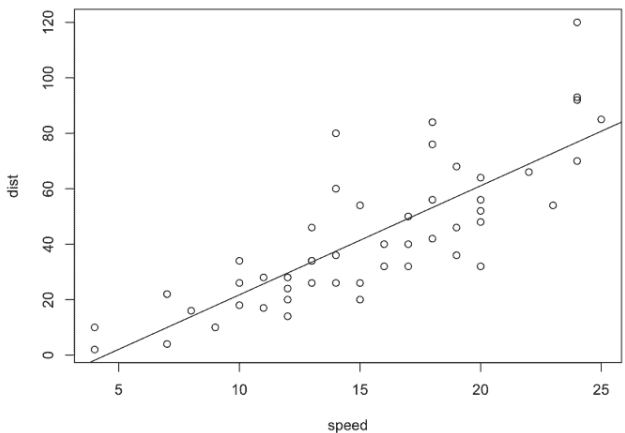
Hello R Markdown

Awesome Me
2018-02-14

This is a paragraph in an R Markdown document.

Below is a code chunk:

```
fit = lm(dist ~ speed, data = cars)
b = coef(fit)
plot(cars)
abline(fit)
```



The slope of the regression is -17.5790949.

R for Data Science:
Chapter 27 R Markdown

R for Data Science:
Chapter 28 Quarto

markdown.qmd

Text formatting

italic **bold** underline ~~strikeout~~ SMALL CAPS code superscript² and subscript₂

Headings

1st Level Header

2nd Level Header

3rd Level Header

Lists

- Bulleted list item 1
- Item 2
 - Item 2a
 - Item 2b

- Numbered list item 1
- Item 2. The numbers are incremented automatically in the output.

Links and images

<http://example.com>

[linked phrase](#)

<http://example.com>



optional caption text

W: 330 h: 77 px 1 Lock ratio

Tables

First Header	Second Header
Content Cell	Content Cell
Content Cell	Content Cell

type to search...

- R Code Chunk
Executable R chunk
- Python Code Chunk
Executable Python chunk
- Div...
Block containing other con...
- Bullet List
List using bullets for items
- Numbered List
List using numbers for items
- H1
Heading 1
Part heading

Data cleaning in action

Can we import this directly into SPSS or R and crunch some numbers?

X4	n	percent	valid_percent
F 2	0.08	0.09523810	
F 1	0.04	0.04761905	
Female 7	0.28	0.33333333	
M 3	0.12	0.14285714	
M 1	0.04	0.04761905	
Male 6	0.24	0.28571429	
Sex 1	0.04	0.04761905	
<NA> 4	0.16	NA	

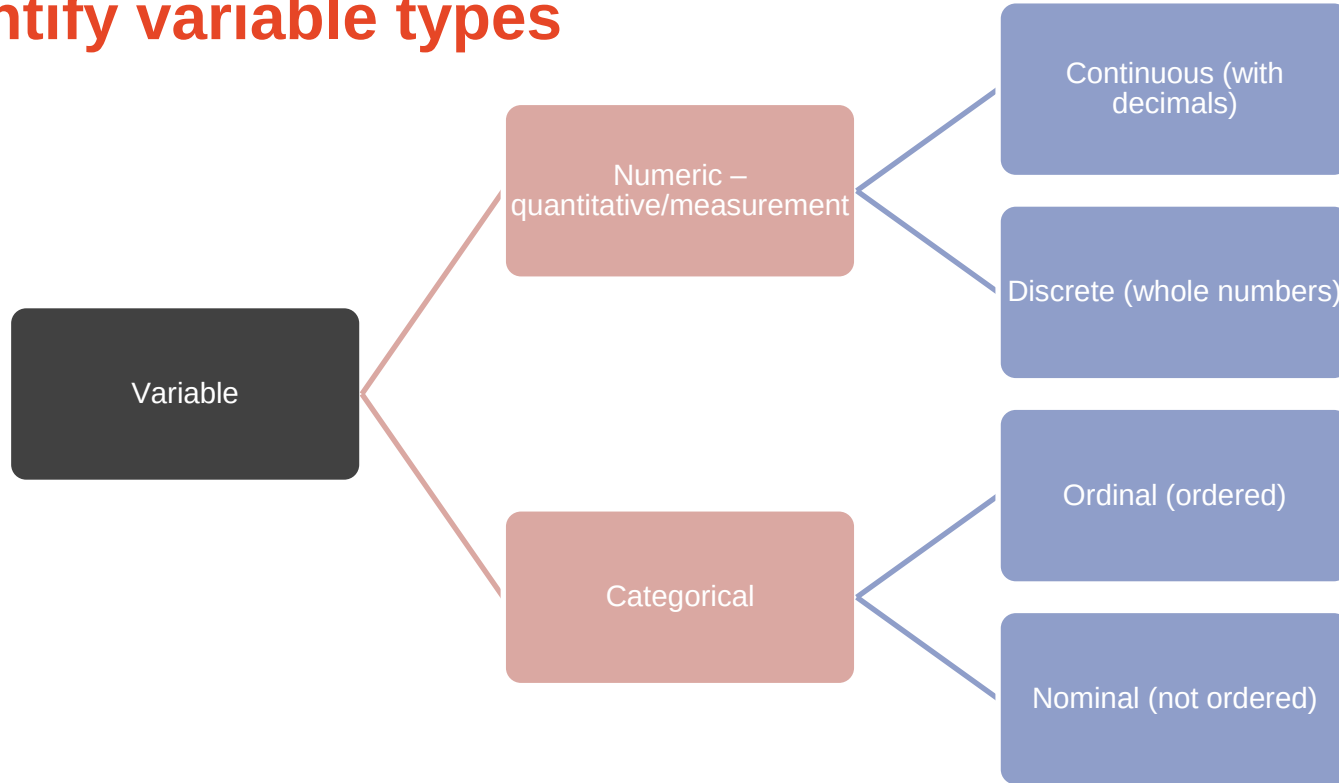
Variable types are important!



Why worry about variable types?

- **Variable types determine the appropriate statistical methods for analysis.**
- You need to know what data type your variable is AND how it is recorded in your data.
- You may need to convert a numeric variable to a categorical variable depending on its distribution.

A. Identify variable types



Variable types

CONTINUOUS

measured data, can have ∞ values within possible range.



I AM 3.1" TALL
I WEIGH 34.16 grams

DISCRETE

OBSERVATIONS can only exist at LIMITED VALUES, often COUNTS.



I HAVE 8 LEGS
and
4 SPOTS!

@allison-horst

Variable types

NOMINAL

UNORDERED DESCRIPTIONS



ORDINAL

ORDERED DESCRIPTIONS



BINARY

ONLY 2 MUTUALLY EXCLUSIVE OUTCOMES



@allison_horst

Variable types – Example Cath dataset

- Dataset from the Duke University Disease Databank – an observational cohort study.
- Patients were referred to the clinic for chest pain and cardiac catheterisation was performed to diagnose and open blockages in the arteries, followed by stenting to keep them open.
- 3504 participants

Cath Dataset Description

Document: This dataset was not published as a manuscript but has been generously provided by Frank Harrell.



Variable types – Example Cath dataset

Codebook

variable	position	Variable_Label	units	Codes	type
sex	1	Gender		0 = male, 1 = female	double
age	2	Age	years		double
cad_dur	3	duration of cardiac symptoms	days		double
choleste	4	Serum Cholesterol	milligrams per deciliter		double
sigdz	5	Significant Coronary Artery Disease Found on Cardiac Cath		0 = no, 1 = yes	double
tvdlm	6	Three Vessel or Left Main Disease Found on Cardiac Cath		0 = no, 1 = yes	double

B. Descriptive analysis for individual variables

Categorical variables

Graphical summaries

- Bar charts

Tabular summaries

- Frequency tables

Numerical variables

Graphical summaries

- Histogram
- Boxplots

Tabular summaries

- Numerical summary statistics of centre (mean, median mode) and spread (quartiles, percentiles, sd)

Example Cath dataset – How to summarise categorical variables?

Cath Dataset Description

Document: This dataset was not published as a manuscript but has been generously provided by Frank Harrell.

Frequency – count the number of males and females

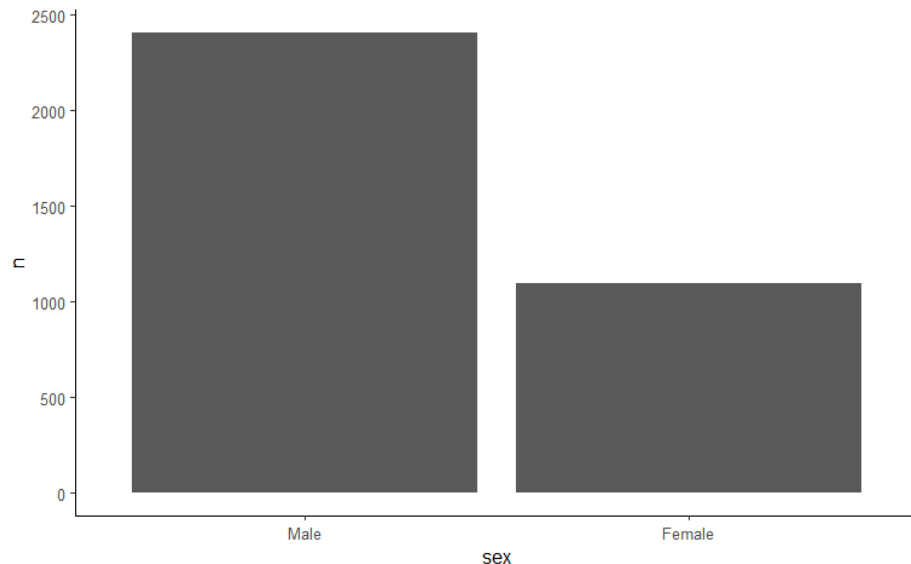
patient_id	sex	age	age [years]	1st Main Disease by Cardiac Cath
1	0	73	132	268
2	0	68	85	120
3	0	54	45	NA
4	1	58	86	245
5	1	56	7	269
6	0	64	0	NA
7	0	65	76	NA
8	0	41	15	247
9	0	68	30	NA
10	0	52	1	NA
11	0	48	1	NA
12	0	35	44	257
13	1	69	10	NA
14	0	58	7	168
15	0	81	2	246
16	0	58	79	221
17	0	59	36	NA
18	0	47	6	272
19	0	66	8	257
20	0	48	69	236

Example Cath dataset – How to summarise categorical variables?

- Frequency – count the number of males and females

```
sex    n    percent
0  2405  0.6863584
1  1099  0.3136416
```

Where 0 = male, and 1
= female



Example Cath dataset – How to summarise numerical variables?

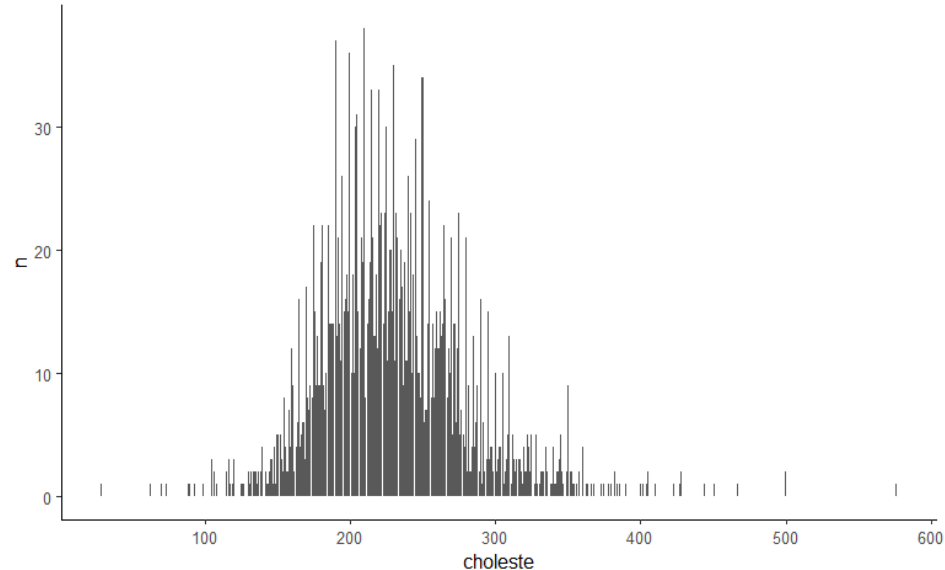
- Serum cholesterol (mg per dl) of the participants who provided a response
- A frequency table would be long and messy. Not a great summary!

	choleste	n	percent	valid_percent
1	29	1	0.0002853881	0.0004428698
2	63	1	0.0002853881	0.0004428698
3	70	1	0.0002853881	0.0004428698
4	74	1	0.0002853881	0.0004428698
5	89	1	0.0002853881	0.0004428698
6	90	1	0.0002853881	0.0004428698
7	93	1	0.0002853881	0.0004428698
8	99	1	0.0002853881	0.0004428698
9	105	3	0.0008561644	0.0013286094
10	107	2	0.0005707763	0.0008857396
11	108	1	0.0002853881	0.0004428698
12	115	2	0.0005707763	0.0008857396
13	117	3	0.0008561644	0.0013286094
14	118	1	0.0002853881	0.0004428698
15	119	1	0.0002853881	0.0004428698
16	120	3	0.0008561644	0.0013286094
17	125	1	0.0002853881	0.0004428698
18	126	1	0.0002853881	0.0004428698
19	127	1	0.0002853881	0.0004428698
20	130	2	0.0005707763	0.0008857396
21	131	1	0.0002853881	0.0004428698
22	132	2	0.0005707763	0.0008857396
23	134	2	0.0005707763	0.0008857396
24	135	2	0.0005707763	0.0008857396
25	136	1	0.0002853881	0.0004428698
26	137	2	0.0005707763	0.0008857396
27	139	2	0.0005707763	0.0008857396
28	140	4	0.0011415525	0.0017714792
29	142	2	0.0005707763	0.0008857396
30	143	1	0.0002853881	0.0004428698
31	144	1	0.0002853881	0.0004428698

	choleste	n	percent	valid_percent
234	353	2	0.0005707763	0.0008857396
235	354	1	0.0002853881	0.0004428698
236	356	1	0.0002853881	0.0004428698
237	358	2	0.0005707763	0.0008857396
238	360	4	0.0011415525	0.0017714792
239	363	1	0.0002853881	0.0004428698
240	364	1	0.0002853881	0.0004428698
241	366	1	0.0002853881	0.0004428698
242	368	1	0.0002853881	0.0004428698
243	373	1	0.0002853881	0.0004428698
244	375	1	0.0002853881	0.0004428698
245	378	1	0.0002853881	0.0004428698
246	380	1	0.0002853881	0.0004428698
247	382	2	0.0005707763	0.0008857396
248	384	1	0.0002853881	0.0004428698
249	386	1	0.0002853881	0.0004428698
250	390	1	0.0002853881	0.0004428698
251	400	1	0.0002853881	0.0004428698
252	402	1	0.0002853881	0.0004428698
253	404	1	0.0002853881	0.0004428698
254	405	2	0.0005707763	0.0008857396
255	410	1	0.0002853881	0.0004428698
256	423	1	0.0002853881	0.0004428698
257	427	1	0.0002853881	0.0004428698
258	428	2	0.0005707763	0.0008857396
259	444	1	0.0002853881	0.0004428698
260	451	1	0.0002853881	0.0004428698
261	467	1	0.0002853881	0.0004428698
262	500	2	0.0005707763	0.0008857396
263	576	1	0.0002853881	0.0004428698
264	N/A	1246	0.3555936073	N/A

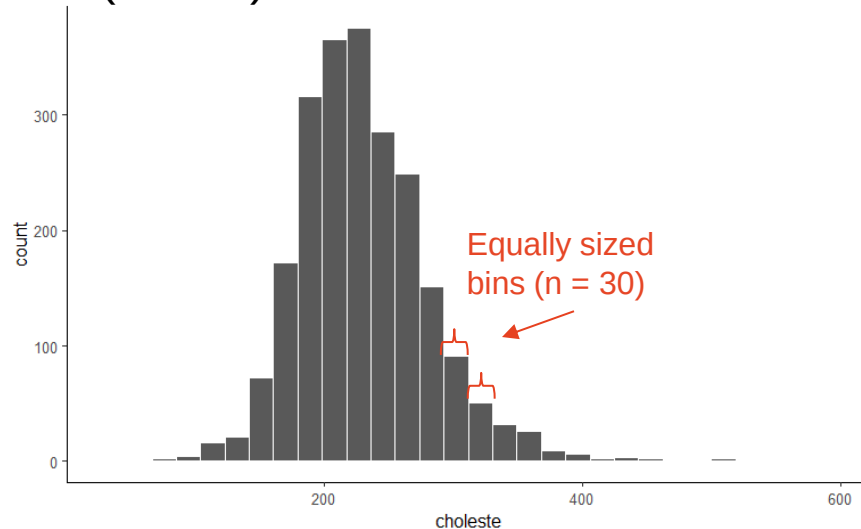
Example Cath dataset – How to summarise numerical variables?

- Serum cholesterol (mg per dl) of the participants who provided a response
- A bar chart isn't the best either! 😞



Example Cath dataset – How to summarise numerical variables?

- **Serum cholesterol (mg per dl) of the participants who provided a response**
- A histogram of cholesterol, with frequency shown with equally sized bins ($n = 30$).





Example Cath dataset – How to summarise numerical variables?

- **Age (years) of the participants who provided a response**
- A frequency table would be long and messy. Not a great summary!

	age	n	percent
1	17	1	0.0002853881
2	18	1	0.0002853881
3	19	1	0.0002853881
4	20	2	0.0005707763
5	22	2	0.0005707763
6	23	1	0.0002853881
7	24	2	0.0005707763
8	25	4	0.0011415525
9	26	2	0.0005707763
10	27	5	0.0014269406
11	28	11	0.0031392694
12	29	9	0.0025684932
13	30	10	0.0028538813
14	31	18	0.0051369863
15	32	17	0.0048515982

age	n	percent
16	33	0.0042808219
17	34	0.0085616438
18	35	0.0085616438
19	36	0.0097031963
20	37	0.0131278539
21	38	0.0159817352
22	39	0.0174086758
23	40	0.0179794521
24	41	0.0239726027
25	42	0.0216894977
26	43	0.0256849315
27	44	0.0291095890
28	45	0.0271118721
29	46	0.0371004566
30	47	0.0316780822

age	n	percent
31	48	0.0382420091
32	49	0.0410958904
33	50	0.0388127854
34	51	0.0353881279
35	52	0.0331050228
36	53	0.0365296804
37	54	0.0359589041
38	55	0.0359589041
39	56	0.0445205479
40	57	0.0410958904
41	58	0.0359589041
42	59	0.0273972603
43	60	0.0282534247
44	61	0.0276826484
45	62	0.0222602740

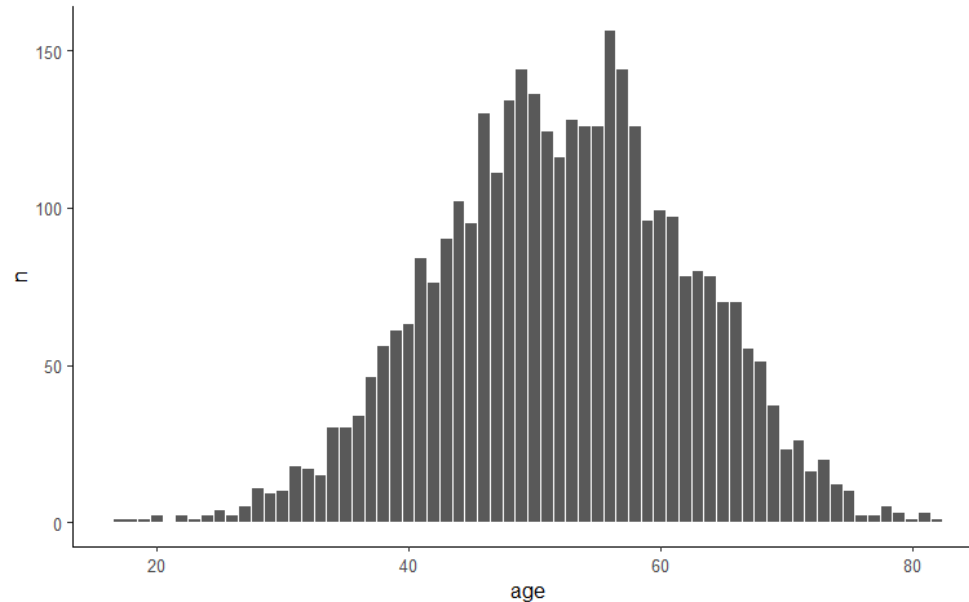
age	n	percent
46	63	0.0228310502
47	64	0.0222602740
48	65	0.0199771689
49	66	0.0199771689
50	67	0.0156963470
51	68	0.0145547945
52	69	0.0105593607
53	70	0.0065639269
54	71	0.0074200913
55	72	0.0045662100
56	73	0.0057077626
57	74	0.0034246575
58	75	0.0028538813
59	76	0.0005707763
60	77	0.0005707763

age	n	percent
61	78	0.0014269406
62	79	0.0008561644
63	80	0.0002853881
64	81	0.0008561644
65	82	0.0002853881



Example Cath dataset – How to summarise numerical variables?

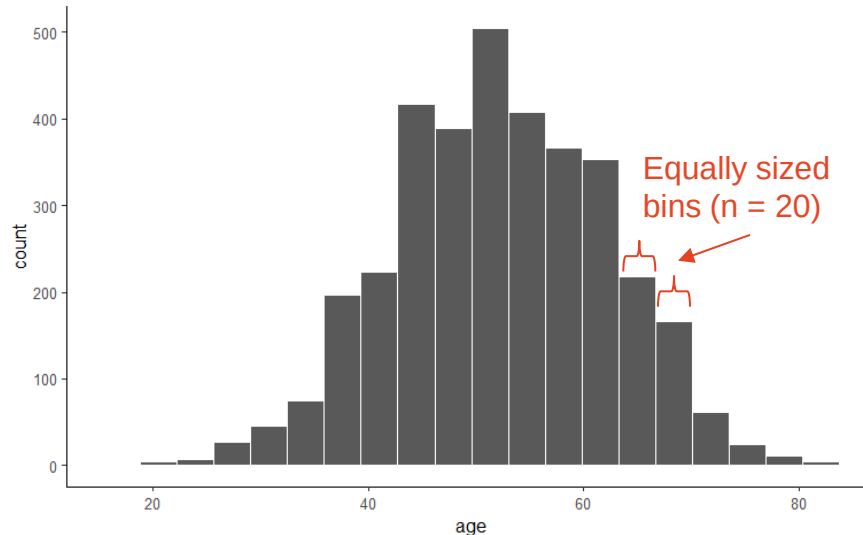
- **Age (years) of the participants who provided a response**
- A bar chart isn't the best either!



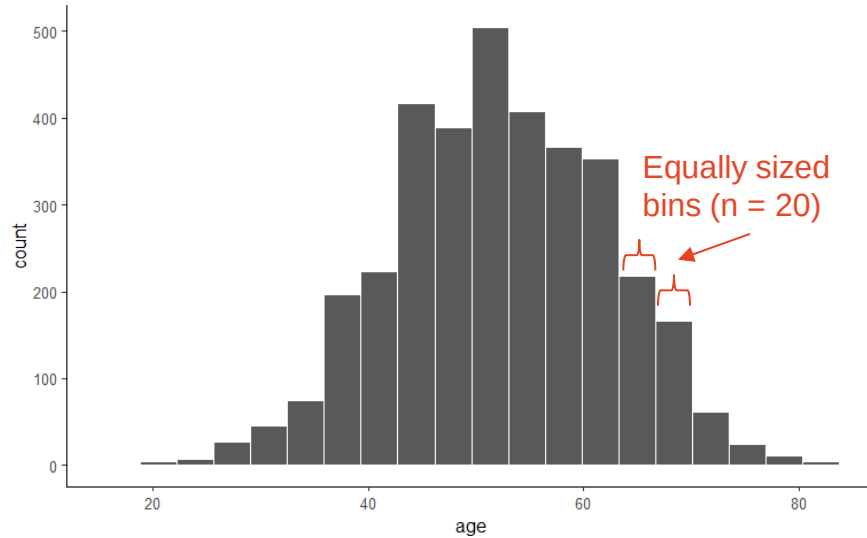


Example Cath dataset – How to summarise numerical variables?

- **Age (years) of the participants who provided a response**
- A histogram of age, with frequency shown with equally sized bins ($n = 20$).

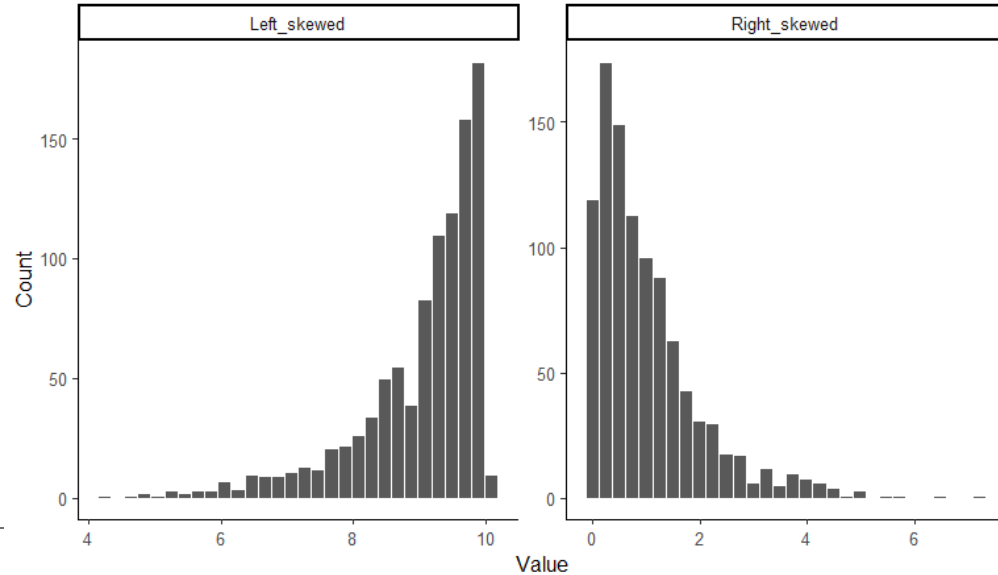


Shapes of the distribution



Symmetric distribution

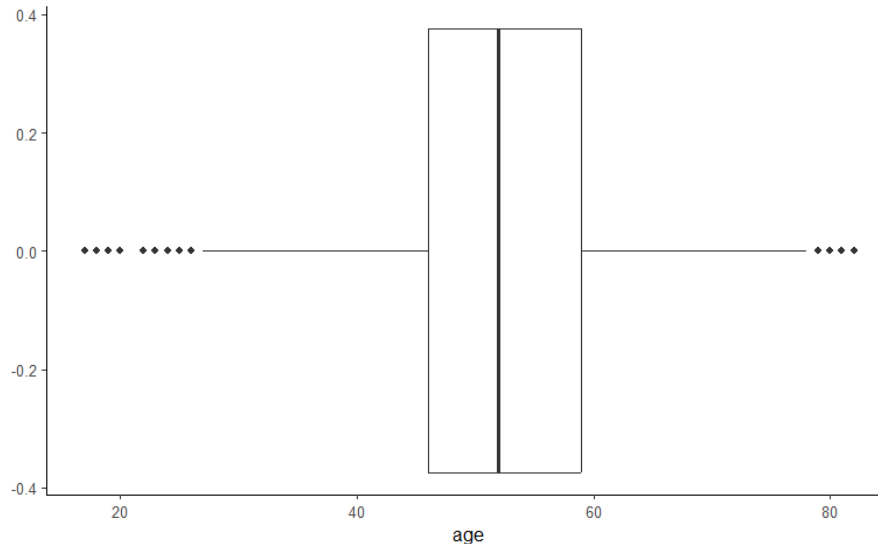
Examples of Skewed Distributions



Asymmetric distribution

Example Cath dataset – How to summarise numerical variables?

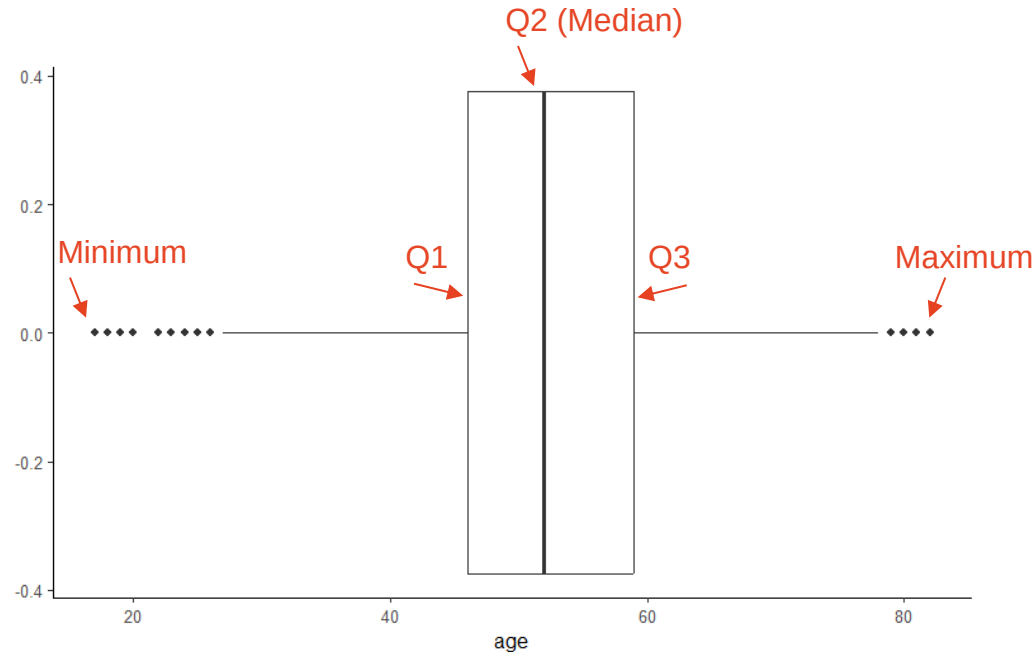
- Another way to summarise a numerical variable is through a boxplot.



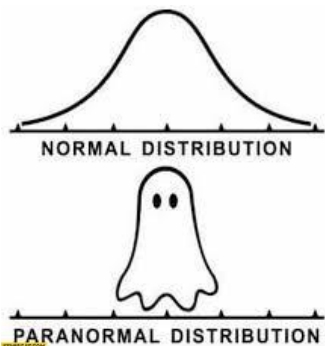
Example Cath dataset – Summarising the boxplot

- A numerical distribution can be summarised by giving descriptions/measures of:

- Its shape
 - Symmetric, right or left skewed
- Its centre
 - Measures of central value or location
- Its spread
 - Measures of spread or dispersion



Parametric versus Nonparametric stats



- **Parametric**

- Based on assumptions about data distribution and shape.
 - Normality
 - Though quite robust in large samples due to the central limit theorem.
 - Homogeneity of variance
- Mean and standard deviation reported.
- Suitable for larger samples, with a normal distribution.
- Examples:
 - 2-sample t-test
 - Generalised linear model (One-way Anova, Least squares regression)
 - Pearson correlation

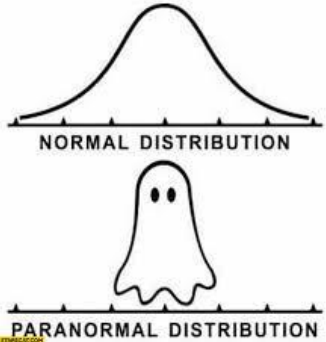
- **Nonparametric**

- Not based on assumptions about data distribution and shape.
- Median and percentiles/quartiles/max and min reported.
- Suitable for smaller samples, skewed data or ordinal variables.
- Examples:
 - Mann-Whitney U
 - Kruskal-Wallis
 - Spearman correlation

With the rise of GLMs and other more flexible parametric methods, such as distributional regression, quantile regression and generalised additive models, nonparametric methods are becoming less useful (and are now less of a requirement for publication).



Improving data distribution with transformations

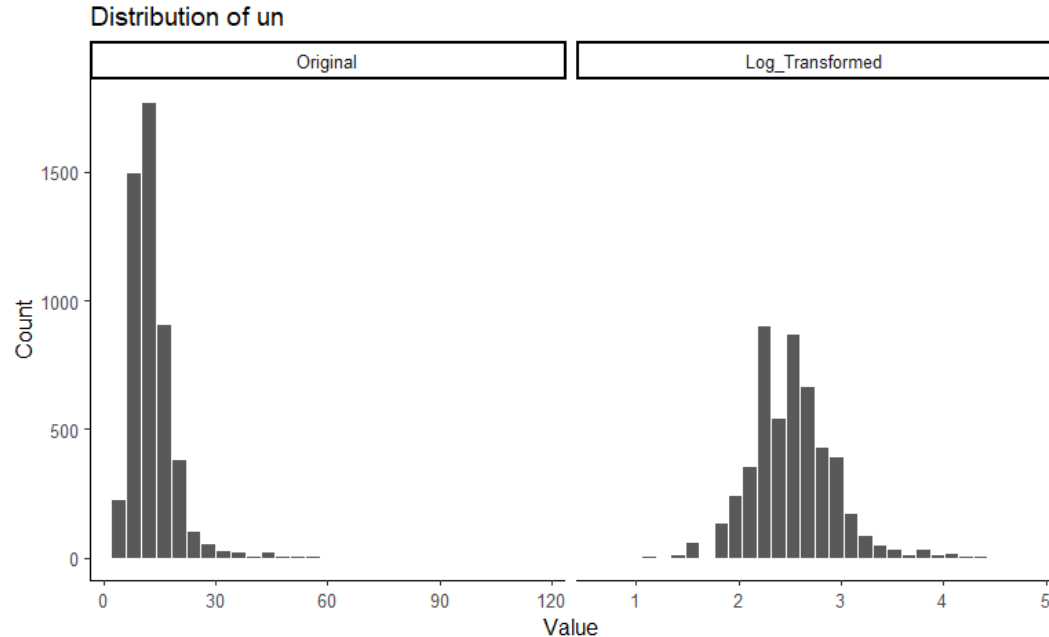


- **Why check the distribution?**
 - Helps us understand skewness and any extreme values in the raw data.
- **Common transformations:** log, square root, inverse, Box-Cox.
 - We cannot take a log of 0, so if the raw data contains values of 0, add a constant to the variable before taking the log. E.g. $\log(x + 1)$.
- Checking the raw data is an important part of EDA. However, when fitting linear models, we assess distributional assumptions using the residuals, not the raw data. See the [Linear Models 1](#) workshop for more information.



Improving data distribution with transformations

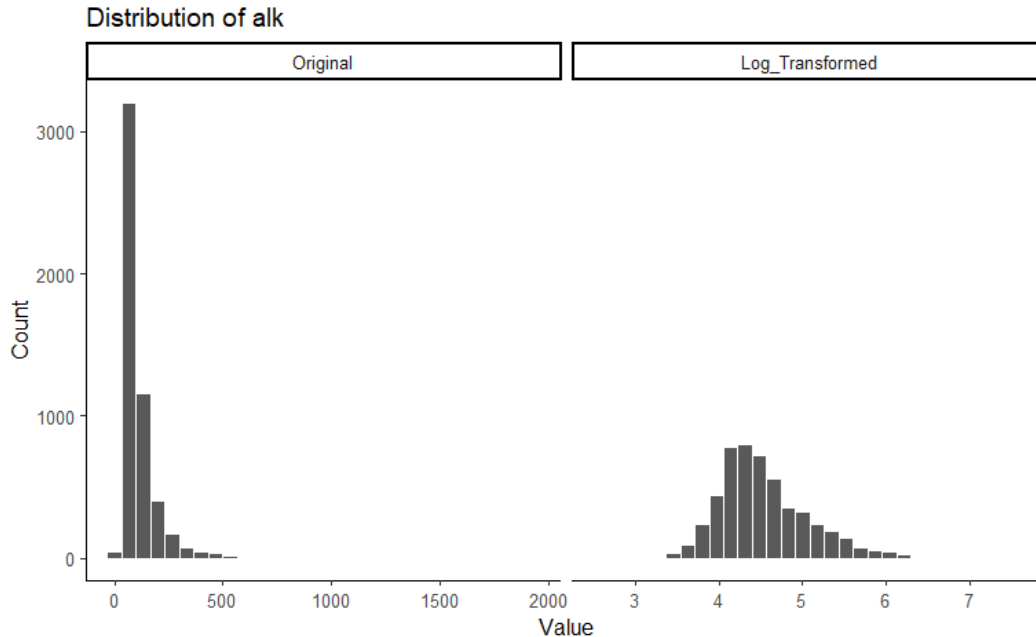
- **UN:** Blood Urea Nitrogen. Numeric. Range: 2-118. 53 missing values.





Improving data distribution with transformations

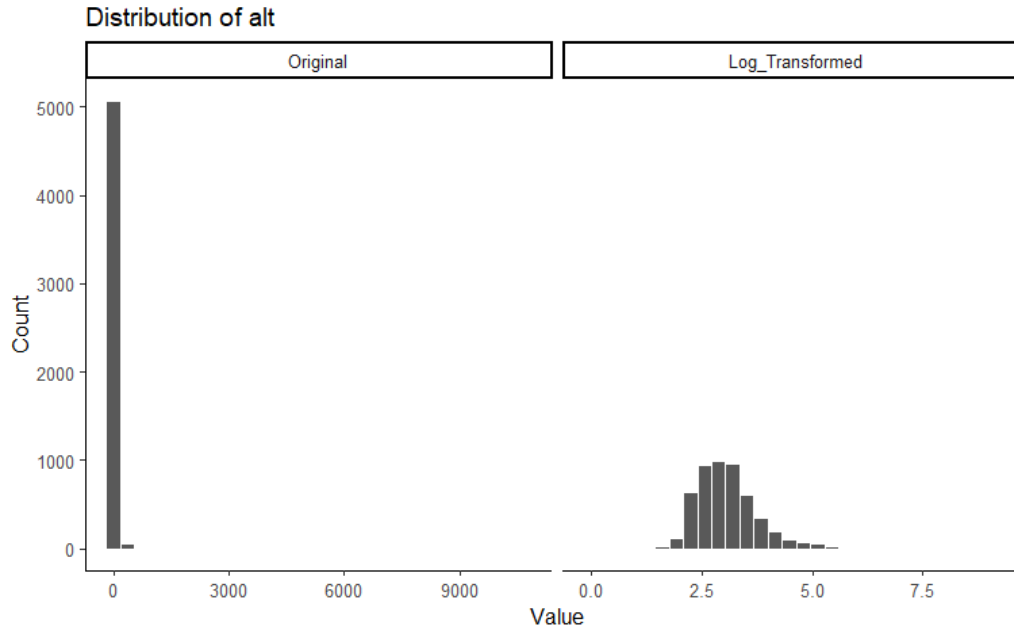
- **ALK:** Alkaline phosphatase. Numeric, range 13-1938, 0 missing values





Improving data distribution with transformations

- **ALT:** Alanine Transaminase. Numeric, range 1-10666, 18 missing values



What is a systematic approach to conduct descriptive analyses for individual variables?

Categorical variables

Graphical summaries

- Bar charts

Tabular summaries

- Frequency tables

Don't forget to check for missing data/NAs!

For formatting tips, check your target journal's instructions to authors or recent publications in that journal!

Numerical variables

Graphical summaries

- Histogram
- Boxplots

Tabular summaries

- Symmetric?
 - Mean, standard deviation, minimum and maximum.
- Asymmetric?
 - Median, quartiles, minimum and maximum.

Variable types are important for describing the distribution of each variable and checking/cleaning each individual variable, but....



What about my research question and analysis?

What is the research question?

Depending on who taught you statistics, your discipline, stats software or textbook, you may have come across some of these terms and more!

- For any analysis we need to be clear what the functional classification of the variables in the data set is, e.g. we want to investigate the effect of smoking on lung disease:



Smoking (yes/no)
Predictor, explanatory
variable, risk factor,
independent variable*



Lung disease (yes/no)
Response, outcome,
dependent variable

*This term is frequently used, but we don't promote it as predictors may be correlated and hence are not independent.

Other functional classifications for variable types

- Covariate: a measured predictor (numeric predictor variable).
- Factor: (a categorical predictor variable).

Experimental design variables:

- Design variables: Based on the physical design of the experiment. They are often included in the analysis even if not 'significant' e.g., correctly partition the variance e.g., Block (batch of reagent, source of lab mice), subject ID, etc.
- Treatment: Variables of interest, e.g., diet, drug treatment, intervention etc. NB: The 'levels' of a 'treatment variable' might include 'control (placebo)', 'treatment 1 (drug 1)', 'treatment 2 (drug 2)'.

More information on design variables in our Experimental Design workshop!

What is the outcome variable?

- **Review study aim and objectives**
- E.g., vaccine RCT - daily morbidity outcome data could be analysed as:
 - mean daily rate (average – numeric).
 - cumulative morbidity (sum – numeric).
 - peak morbidity (maximum – numeric).
 - outbreak presence/absence (binary group – categorical).
 - time to infection/disease outbreak (time to binary event data – survival analysis).

More data processing

- **Assess all variables for missing observations – if many missing consider analysing with and without that predictor.**
- **Check the distribution of all variables individually (previous step).**
- Continuous predictors: handle as continuous or categorical?
- Categorical: may have to combine categories if there are low frequency counts (if it makes sense to do so).
- **Multi-level (clustered) data**
- Each observation/row uniquely identified? E.g., herd, animal, ID.
- Evaluate hierarchical structure of your data: Average/range of observations at one level in each higher level?
- E.g., mean, min, max of students/class; mean, min, max of classes/school.

A quick primer to Step 5: EDA

Step 5: Exploratory data analysis (EDA)



It will depend on the analysis and variables involved.



Basic EDA is where you plot the relationship of each predictor with the outcome, and you may need to reconsider data processing. E.g. Do I need to merge more categories together?

Two categorical variables

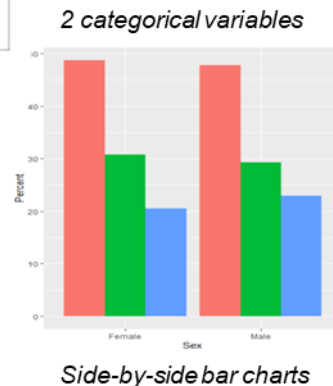
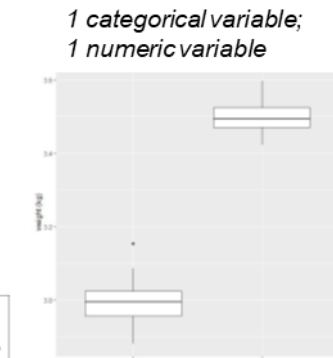
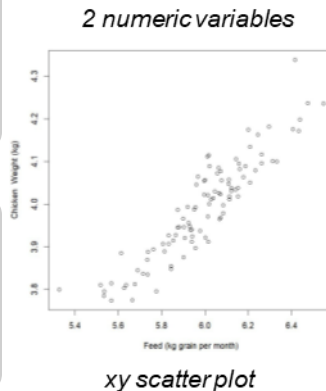
- Contingency table
- Side-by-side bar chart

A categorical and a numeric variable

- Tabulate summary statistics by groups
- Box-and-whisker plot by groups

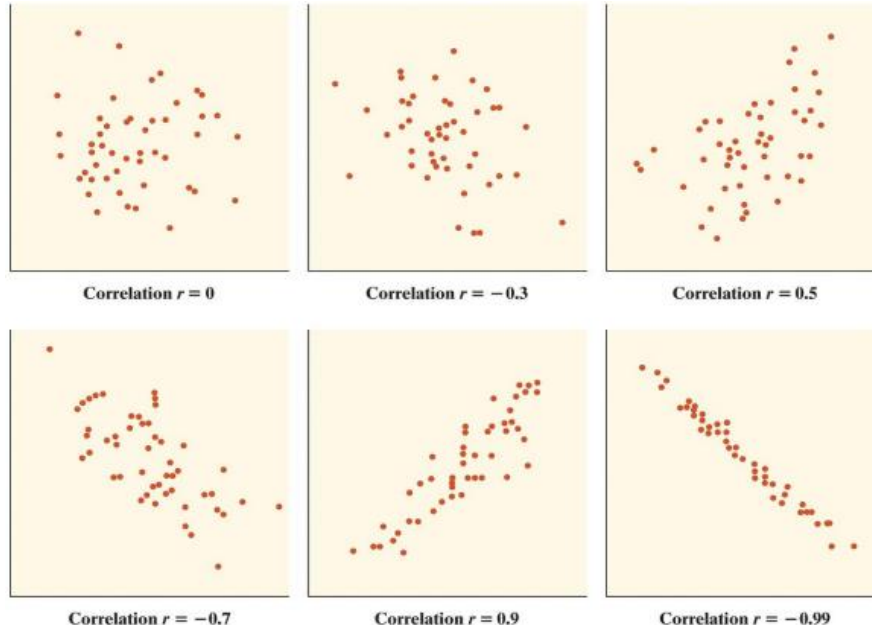
Two numeric variables

- Scatter plot and correlation coefficient r



XY scatterplot and Pearson's correlation coefficient (r)

A quick review of correlation coefficient r to describe the relationship of two numeric variables in a scatter plot; $r = 0$ means no relationship – data points in a horizontal line.



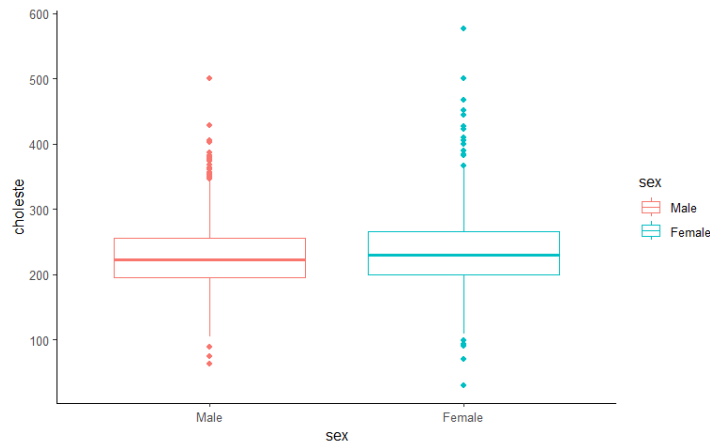
Examples for reporting descriptive analyses:

- Plotting gives a quick visual summary during EDA and highlights any issues.
- Tables are more publication friendly as they save space!
 - This step of the statistical analysis process is often seen in Table 1 of your manuscript.
 - Who is in this study, anyway? Guidelines for a useful Table 1.

Examples for reporting descriptive analyses: Cath dataset

Table 1. Summary statistics for the variables in a study of patients undergoing cardiac catheterisation, split according to sex.

	Male (N=2405)	Female (N=1099)	Overall (N=3504)
Age [Year]			
Mean (SD)	51.6 (9.83)	53.8 (9.99)	52.3 (9.93)
Median [Min, Max]	52.0 [17.0, 82.0]	54.0 [20.0, 81.0]	52.0 [17.0, 82.0]
Duration of Symptoms of Coronary Artery Disease			
Mean (SD)	43.0 (58.1)	43.0 (58.3)	43.0 (58.2)
Median [Min, Max]	17.0 [0, 416]	20.0 [0, 404]	18.0 [0, 416]
Cholesterol [mg %]			
Mean (SD)	227 (47.3)	237 (56.9)	230 (50.6)
Median [Min, Max]	223 [63.0, 500]	230 [29.0, 576]	225 [29.0, 576]
Missing	836 (34.8%)	410 (37.3%)	1246 (35.6%)
Significant Coronary Artery Disease			
No	533 (22.2%)	637 (58.0%)	1170 (33.4%)
Yes	1872 (77.8%)	462 (42.0%)	2334 (66.6%)
Three Vessel or Left Main Disease			
No	1463 (60.8%)	909 (82.7%)	2372 (67.7%)
Yes	941 (39.1%)	188 (17.1%)	1129 (32.2%)
Missing	1 (0.0%)	2 (0.2%)	3 (0.1%)



[Cath Dataset Description Document](#): This dataset was not published as a manuscript but has been generously provided by Frank Harrell.

Tips from the consulting room



- If you're planning to publish your work, be sure to review the journal's "Instructions for authors" and browse recent articles to guide your layout and presentation!



- Consider who will be reading and viewing your work and present your findings in a way that's easy for them to understand.
 - Is it clinicians? Is it industry? Is it other researchers? Is it a broad audience/lay people?

Best-practice reporting guidelines

- [Equator network](#) (Enhancing the QUALity and Transparency Of health Research).
 - Seeks to improve the reliability and value of published health research by promoting transparent and accurate reporting.
- There are guidelines for different study types to help you with your manuscript writing.
 - Provides you with a check-list of information required so that your manuscript can be easily understood by the reader and reproduced by other researchers.



Reporting guidelines for main study types

Randomised trials	CONSORT	Extensions
Observational studies	STROBE	Extensions
Systematic reviews	PRISMA	Extensions
Study protocols	SPIRIT	PRISMA-P
Diagnostic/prognostic studies	STARD	TRIPOD
Case reports	CARE	Extensions
Clinical practice guidelines	AGREE	RIGHT
Qualitative research	SRQR	COREQ
Animal pre-clinical studies	ARRIVE	
Quality improvement studies	SQUIRE	Extensions
Economic evaluations	CHEERS	Extensions





Statistical analysis plans (SAPs)



- SAPs as well as study protocols are a great tool to improve communication and collaboration between you and other researchers.
- They should include information on:
 - Objectives and hypotheses
 - Primary and secondary outcomes
 - Study design
 - Data collection methods
 - Data management
 - Statistical analysis including handling of missing data and sensitivity analyses
 - Reporting of findings
- [Australian Clinical Trials Alliance: Statistical analysis plan](#)
- [Jama: Guidelines for statistical analysis plans](#)
- [A template for the authoring of statistical analysis plans](#)
- [Developing a Quantitative Data Analysis Plan for Observational Studies](#)

Some analysis examples

5. Exploratory data analysis (EDA)

6. Inferential analysis

Data analysis workflow: 4 examples

- **A.** Linear models examples – Simple regression, ANOVA, ANCOVA, Repeated measures.
- **B.** Extended linear models example – Survival analysis.
- **C.** Extended linear models example – Generalised linear model (Binary logistic regression)
- **D.** Multivariate analysis – Confirmatory factor analysis (CFA)

Example A: Linear models examples

Scenario: We are interested in studying a continuous outcome variable, e.g. weight gain (kg) or blood cell count (cells/ μ L).

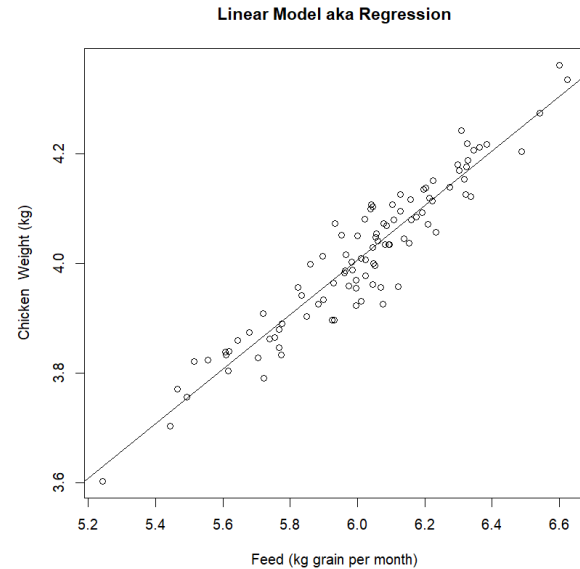
- 1. Simple linear regression – one continuous predictor variable.
- 2. ANOVA (control vs. treatment) – for 2 groups = 2 sample t-test = simple linear regression with one binary predictor variable.
- 3. ANCOVA – ANOVA with a covariate predictor.

For more detail on how to do these analyses including R code, attend our SIH [Linear Models 1: Linear Regression, ANOVA, ANCOVA and Repeated Measures \(a Simple Mixed Model\)](#) workshop!

Example A1: Linear models – Simple linear regression

Step 5: EDA – Plot the data in a scatterplot.

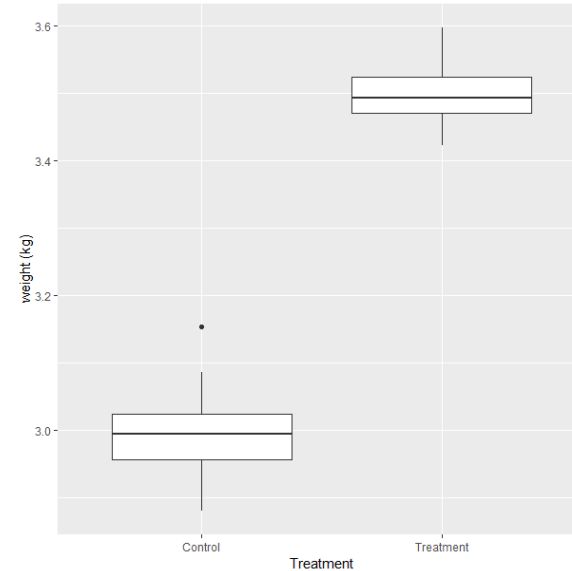
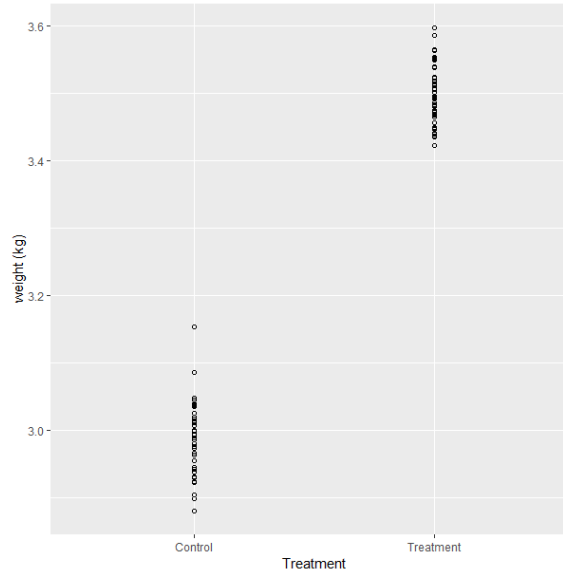
Step 6: Inferential analysis – fit a linear regression line and test if the slope is different from 0; $P < 0.001$; report slope/regression estimate and 95% CI.



Example A2: Linear models – Control vs. treatment experiment

Step 5: EDA – Plot the data with grouped boxplots.

Step 6: Inferential analysis – ANOVA/2-sample t-test; $P < 0.001$; report predicted means and 95% CIs.

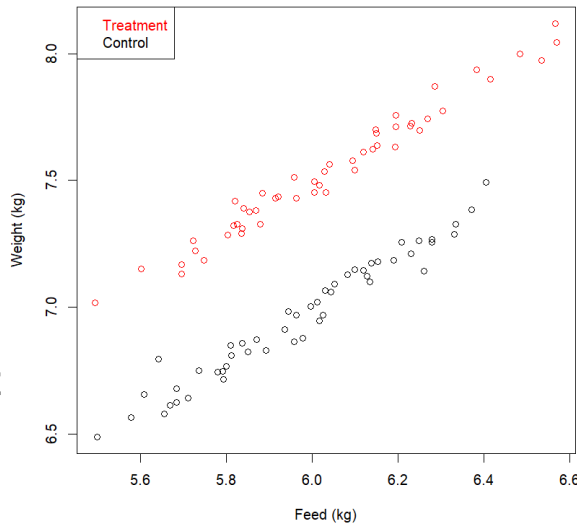


Example A3: Linear models – ANCOVA (ANOVA with a continuous covariate)

Step 5: EDA – Plot the data (differentiate categories of the treatment variable).

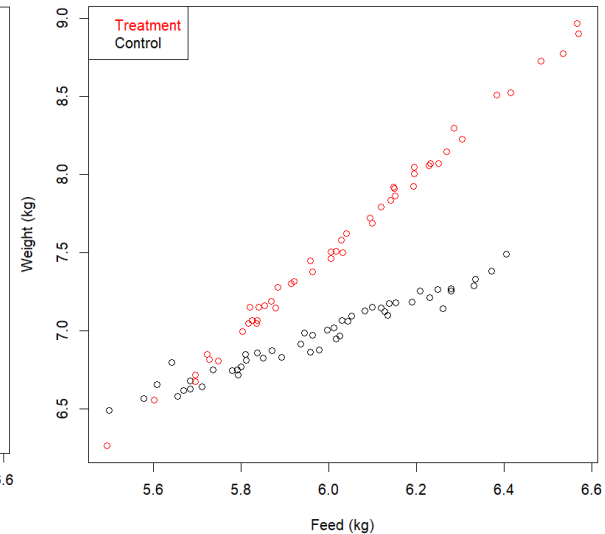
Step 6: Inferential analysis – ANCOVA or multivariable regression; $P < 0.001$; report predicted means with or without interaction and 95% CIs.

Supplement has no impact on feed's relationship with weight



Without interaction

Supplement has an impact on feed's relationship with weight



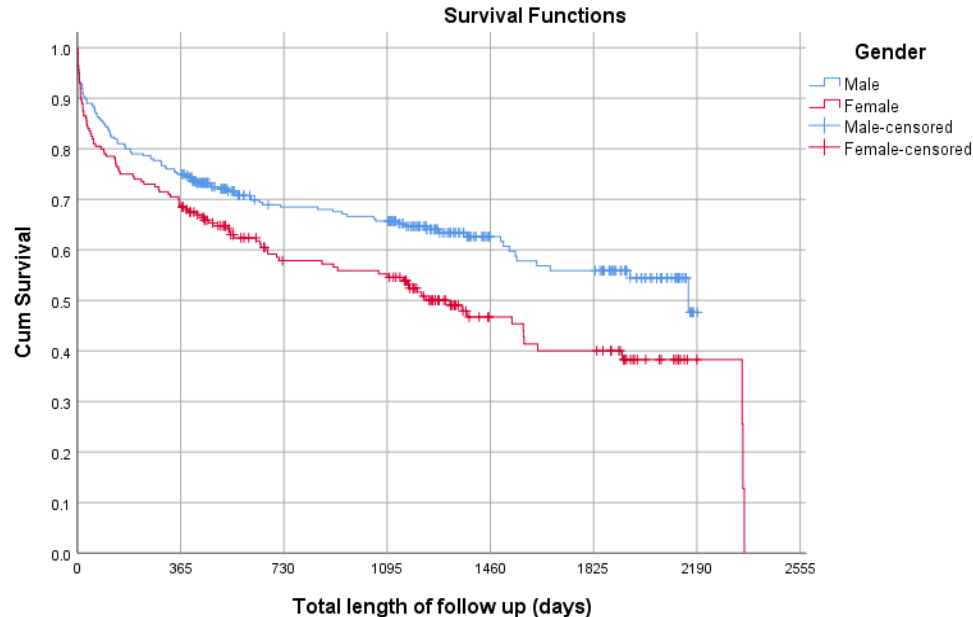
With interaction

Example B: Survival analysis

- **Scenario:** Worcester Heart Attack Community Surveillance Study (WHAS)
- **Aim:** To examine time trends in the incidence rate of acute heart attacks.
- **Objective:** Investigate if different demographic and clinical factors are associated with the time to a heart attack.
- **Data:** Longitudinal, observational data.
- **Outcome:** Heart attack – time to event.
- **Predictors:** Demographic and clinical data.
- **Key feature:** Data is censored – see our Introduction to Survival Analysis WS.

Example B: Survival analysis

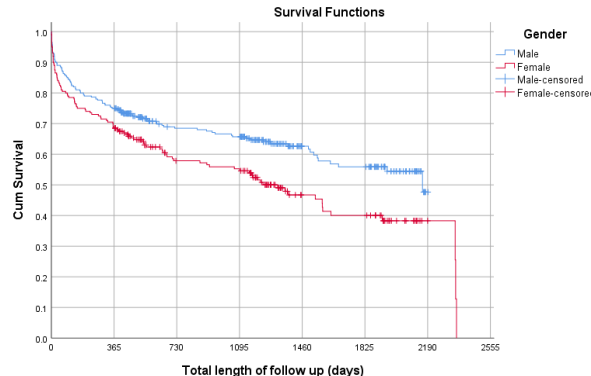
Step 5: EDA – Kaplan Meier curve is the EDA plot for survival analysis.



Example B: Survival analysis

Step 6: Inferential analysis – The logrank test to compare the survival curves of two or more groups.

- There is a significant difference in survival between males and females (by log-rank test).
 - Median survival for males: 2160 days [95%CI: not calc].
 - Median survival for females: 1317 days [95% CI 970-1664].



Example C: Binary logistic regression

- **Scenario:** Survey responses of 32 long-term smokers to determine the association between smoking status and lung-cancer diagnosis.
- **Outcome:** Lung cancer (no/yes)
- **Predictors:** Years smoking, BMI
- [Primer on binary logistic regression](#)

For more detail on how to do these analyses including R code, attend our SIH [Linear Models 2: Logistic and Poisson/Count Regression - An Introduction to Generalised Linear Models](#) workshop!

Example C: Binary logistic regression

Step 5: EDA – Log odds of the outcome (lung cancer) vs. numerical predictor (years of smoking) to check the linearity assumption.

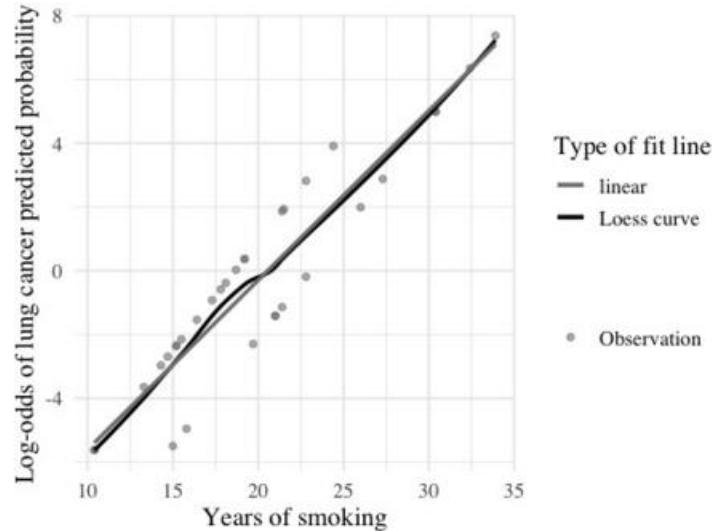


Figure 2

Checking the linearity assumption graphically.

Example C: Binary logistic regression

Step 6: Inferential analysis – binary logistic regression to examine association between lung cancer (y/n) and years smoking (numerical), if linearity assumption is met.

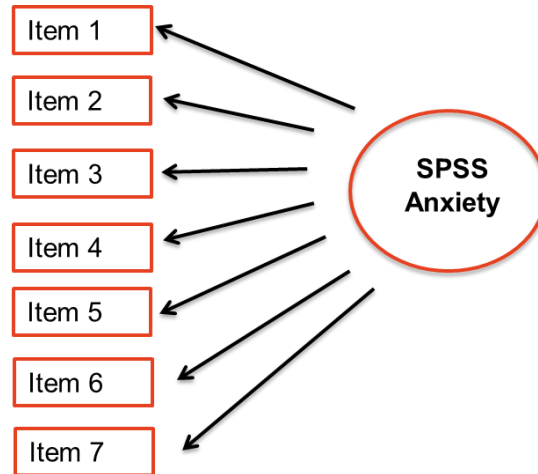
For every additional year of smoking, the odds of lung cancer are approximately 1.98 times higher (95% CI 1.36 to 3.86).

Example D: Multivariate analysis – Confirmatory factor analysis

- **Scenario:** To test if a factor model 'SPSS statistical software Anxiety' explains the common variance among 7 questionnaire items:
 1. I dream that Pearson is attacking me with correlation coefficients.
 2. I have little experience with computers.
 3. All computers hate me.
 4. I have never been good at mathematics.
 5. My friends are better at statistics than me.
 6. Computers are useful only for playing games.
 7. I did badly at mathematics at school.

Example D: Multivariate analysis – Confirmatory factor analysis

- **Scenario:** Confirm SPSS Anxiety as a factor explaining the common variance among the 7 items.



Example adapted from [A Practical Introduction to Factor Analysis: Confirmatory Factor Analysis](#) from the UCLA: Statistical Consulting Group.

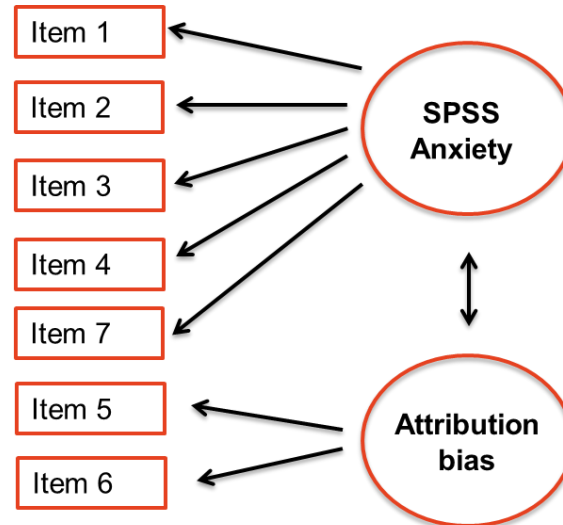
Example D: Multivariate analysis – Confirmatory factor analysis

Step 5: EDA – scatter plots + Pearson's correlation coefficient r ; correlation matrix.

	Q1	Q2	Q3	Q4	Q5	Q6	Q7
Q1	1						
Q2	-0.34	1					
Q3	0.44	-0.38	1				
Q4	0.40	-0.31	0.40	1			
Q5	0.22	-0.23	0.28	0.26	1		
Q6	0.31	-0.38	0.41	0.34	0.51	1	
Q7	0.33	-0.26	0.35	0.27	0.22	0.30	1

Example D: Multivariate analysis – Confirmatory factor analysis

Step 6: Inferential analysis – to test if a 2-factor model ‘SPSS statistical software Anxiety’ and ‘Attribution bias’ explains the common variance among 7 questionnaire items.





Final notes on Step 6: Inferential analysis

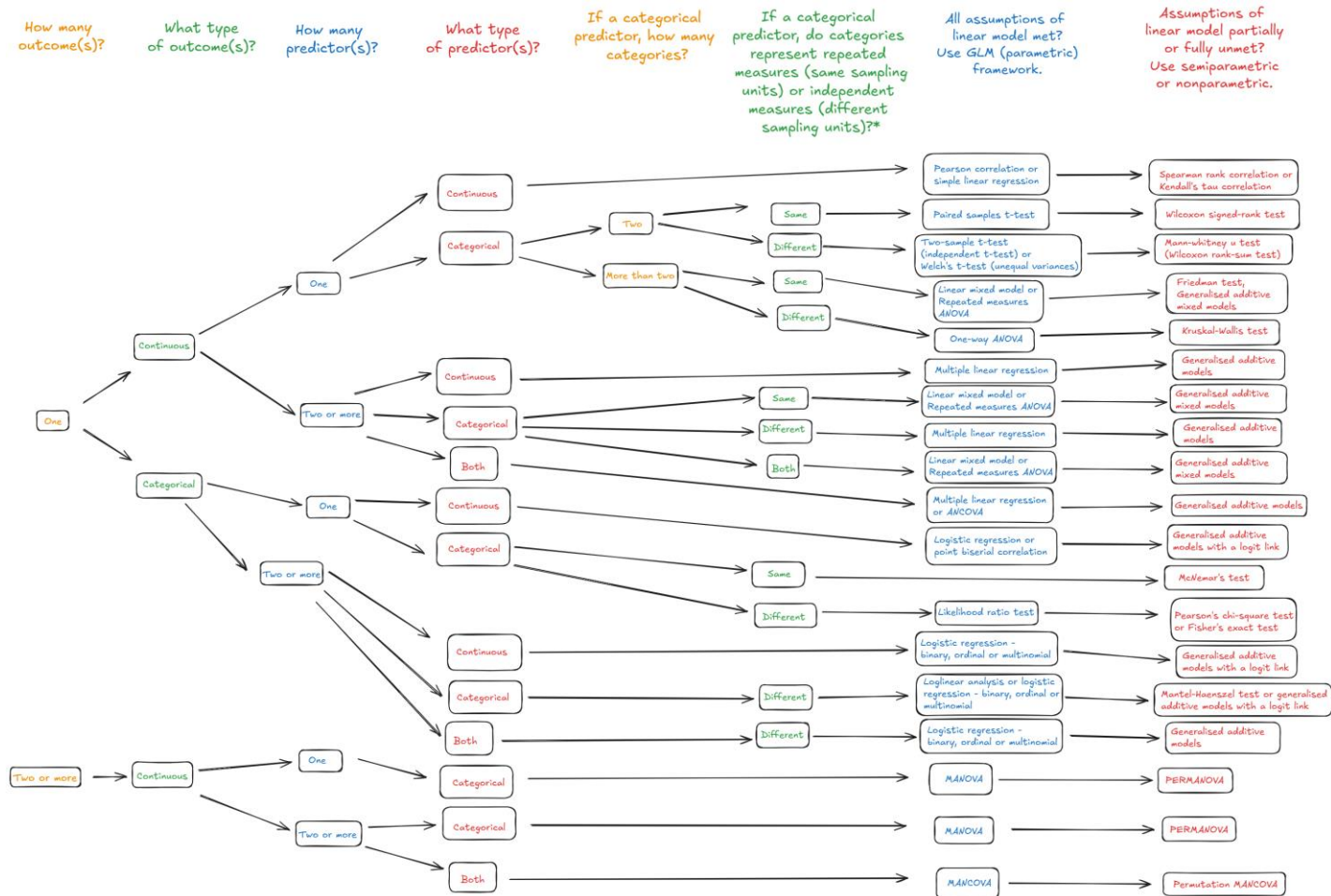
- **We only showed some common examples of statistical analyses – there are many different types of analyses you can consider:**
 - Other Linear Models extensions such as Poisson regression and more complex mixed models - see our SIH [Linear Models 2: Logistic and Poisson/Count Regression - An Introduction to Generalised Linear Models](#) workshop.
 - Survival Analysis for ‘time-to-event’ outcome data – see our SIH [Introduction to Survival Analysis](#) training!
 - Survey Data analysis – see our [Design and Analysis of Surveys 1](#) and [Design and Analysis of Surveys 2: Advanced Topics](#).
 - Other Multivariate Analyses – for example PCA, Factor Analysis - see our [Multivariate Statistical Analysis 1: Dimension Reduction](#).
 - And more [more workshops](#)!



Final notes on Step 6: Inferential analysis

- Start simple and increase complexity step-by-step.
- Always consider/check the test/model assumptions.
- Report 95% CIs for estimates, e.g., predicted means/probabilities/rates.
- For basic analyses consider more powerful (parametric) analyses first and use less powerful tests if assumptions are violated. e.g.:
 - Use a 2-sample t-test with equal or unequal variance for means, before a Mann-Whitney test.
 - Use a chi-squared test to compare proportions before a Fisher's exact test.
- Use knowledge of variable types to guide you through the systematic roadmap on the next slide!

Statistical inferential analysis roadmap



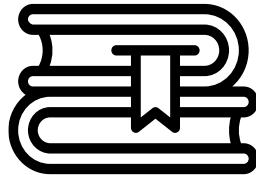
Where possible, if the assumptions are met, use the parametric test as it has more statistical power to detect differences than the nonparametric test equivalent.

*If you are unsure of the units in your study, see our [Experimental Design](#) workshop for more information!



Further R resources

- There is a large online community of R users contributing to free ‘packages’ with data analysis functions, which leads to many ways of coding your analysis in R. This can be confusing. We recommend using tidyverse packages and tidy-centric code.
- See our SIH helpful links for guides on using [R and Rstudio](#).
- [LinkedIn Learning: R courses](#)
 - Including [Learning the R Tidyverse \(2024\)](#), [Complete Guide to R: Wrangling, Visualizing, and Modelling Data](#), and [Cleaning Bad Data in R](#).
- [RLadiesSydney: RYouWithMe](#)



Other resources

Books on R

- [R for Data Science](#) by Hadley Wickham.

Statistical blogs and websites

- [The Analysis Factor](#)
 - [Seven Steps for Data Cleaning](#)
 - [Best Practices for Organizing your Data Analysis](#)
 - [Best Practices for Data Preparation](#)
 - [Preparing Data for Analysis is \(more than\) Half the Battle](#)
 - [Four Weeds of Data Analysis That are Easy to Get Lost In](#)

Further assistance at The University of Sydney



SIH

- [Statistical Resources](#) website: containing our workshop slides and our favourite external resources (including links for learning R and SPSS).
- [Hacky Hour](#): an informal monthly meetup for getting help with coding or using statistics software.
- 1on1 Consults can be requested on our website or [here](#) (click on the big red 'contact us' link).

SIH Workshops

- Create your own custom programs tailored to your research needs by attending more of our Statistical Consulting workshops. Look for the statistics workshops on our training page or on our [Training calendar](#).
- Sign up to our mailing list to be notified of upcoming training.

Other

- Open Learning Environment (OLE) courses
- LinkedIn Learning

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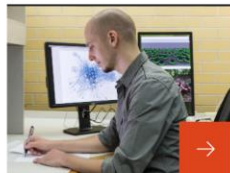
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Workshops and training



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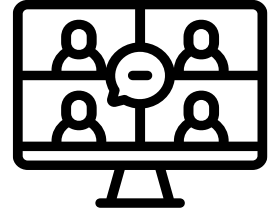
And fill out our request form

Or send us an email:



sih.info@sydney.edu.au

How to use our workshops



- Workshops developed by the Statistical Consulting Team within the Sydney Informatics Hub form an integrated modular framework. Researchers are encouraged to choose modules to **create custom programs tailored to their specific needs**. This is achieved through:
 - Short 90-minute workshops, acknowledging researchers rarely have time for long multi day workshops.
 - Providing statistical workflows applicable in any software, that give practical step by step instructions which researchers return to when analysing and interpreting their data or designing their study e.g. workflows for designing studies for strong causal inference, model diagnostics, interpretation and presentation of results.
 - Each one focusing on a specific statistical method while also integrating and referencing the others to give a holistic understanding of how data can be transformed into knowledge from a statistical perspective from hypothesis generation to publication.

For other workshops that fit into this integrated framework, refer to our training link page under statistics, found below:

[Workshops and training](#)



Online statistical consulting resources



sydney.edu.au/informatics-hub

Useful links

Use the links below to access your tool or platform quickly.

[DashR \(researcher dashboard\)](#) →

[eNotebooks](#) ✓

[REDCap](#) →

[University Code Repository \(GitHub\)](#) →

[Research computing user guide](#) ✓

[SIH PPMS login](#) ✓

[View all tools and platforms](#) →

[SIH Rates for project support](#) ↓

[Statistical consulting resources](#) ✓

Sites

[Workshops and Workflows](#)

The Statistical Consulting Unit

[About Us](#)

[Contact Us](#)

[What to expect in a consult](#)

[Staff](#)

[Acknowledgements](#)

References

[Basic Statistical Theory](#)

[Linear Models](#)

[Logistic Regression](#)

[Meta Analysis](#)

Workshops and Workflows ~ download all our workshops with their step-by-step workflows on how to do analysis.

References ~ 1-3 Curated links on a variety of common statistical concepts and tests e.g.. Basic Theory, Meta Analysis, Software, etc. ***A great place to start looking for help!!***

Statistical Consulting Unit/What to expect in a consult ~ for tips on how to get the most out of your consult.

(You may need to click “Expand all” on the Sites sidebar to get an overview of the available pages).

We recommend our experimental design and sample size workshops

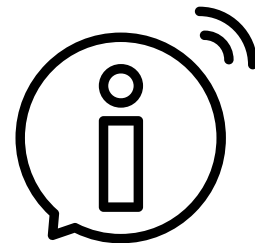
Experimental design workshop

- Far too many researchers think they know all they need to in this area. We commonly see designs that could be substantially improved for stronger causal inference and improved results which leads to publication in higher impact journals (amongst other benefits).
- Even if you have already collected your data, it is well worth attending since it may improve your write up and analysis. E.g., we had a client who didn't realise they had a very strong before/after control/impact (BACI) design.

Power and sample size workshop

- Shows the steps and decisions researchers need to make when designing an experiments to ensure sufficient sample e.g., power, minimum sample required to fit the necessary model, etc.
- Also, how much power the study has, i.e., does it have sufficient power to detect the effects you expect to see, or is your study a complete waste of time and resources?

A reminder: Acknowledging SIH



- All University of Sydney resources are available to researchers free of charge.
- The use of the SIH services including the Artemis HPC and associated support and training warrants acknowledgement in any publications, conference proceedings or posters describing work facilitated by these services.
- The continued acknowledgment of the use of SIH facilities ensures the sustainability of our services.

Suggested wording for use of workshops and workflows:

- *“The authors acknowledge the Statistical workshops and workflows provided by the Sydney Informatics Hub, a Core Research Facility of the University of Sydney.”*

We value your feedback



- We want to hear about you and whether this workshop has helped you in your research. What worked and what didn't work.
- We actively use the feedback to improve our workshops.
- Completing this survey really does help us and we would appreciate your help! It only takes a few minutes to complete (promise!)
- You will receive a link to the anonymous survey by email.



Appendix: Data analysis – Some terminology

- **Univariate/univariable**
– involving one variable, e.g. one outcome per analysis; analysis with one predictor variable.
- **Multivariate** – multiple outcomes in the same analysis/model.
- **Multivariable** – multiple explanatory variables in the same analysis/model.
- **Linear models** (LM – continuous outcome).
- **Generalised linear models** (GLM – categorical outcomes, e.g. binary, ordinal, multinomial (for nominal outcome data) or Poisson/negative binomial regression (for count/rate outcome data)).
- **Mixed models** (i.e. LM or GLM with a random effect = LMM or GLMM) where data are clustered in space or time, e.g. repeated measures/longitudinal data.